



Big Data
Lab 6 – Logistic Regression

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Question1

What are the variable pairs that are not correlated at all to each other?

from the correlation matrix the pairs that are not correlated are (Price, Income) and (Price, Age)

	Price	Income	Age
Price	1	0.00000000	0.00000000
Income	0	1.00000000	0.09612083
Age	0	0.09612083	1.00000000

Question2

Are there any highly correlated variables in this dataset?
No, there is a weak correlation between (Age, Income) but this is not high and there is no other correlation.

Question3

How many categories are there for the Price variable?

3 Categories (10,20,30)

Question4

Why is it divided into two entries only in the model?

Because logistic regression uses dummy variables, one category is taken as a reference variable.
So if there are 3 categories only 2 variables are created.

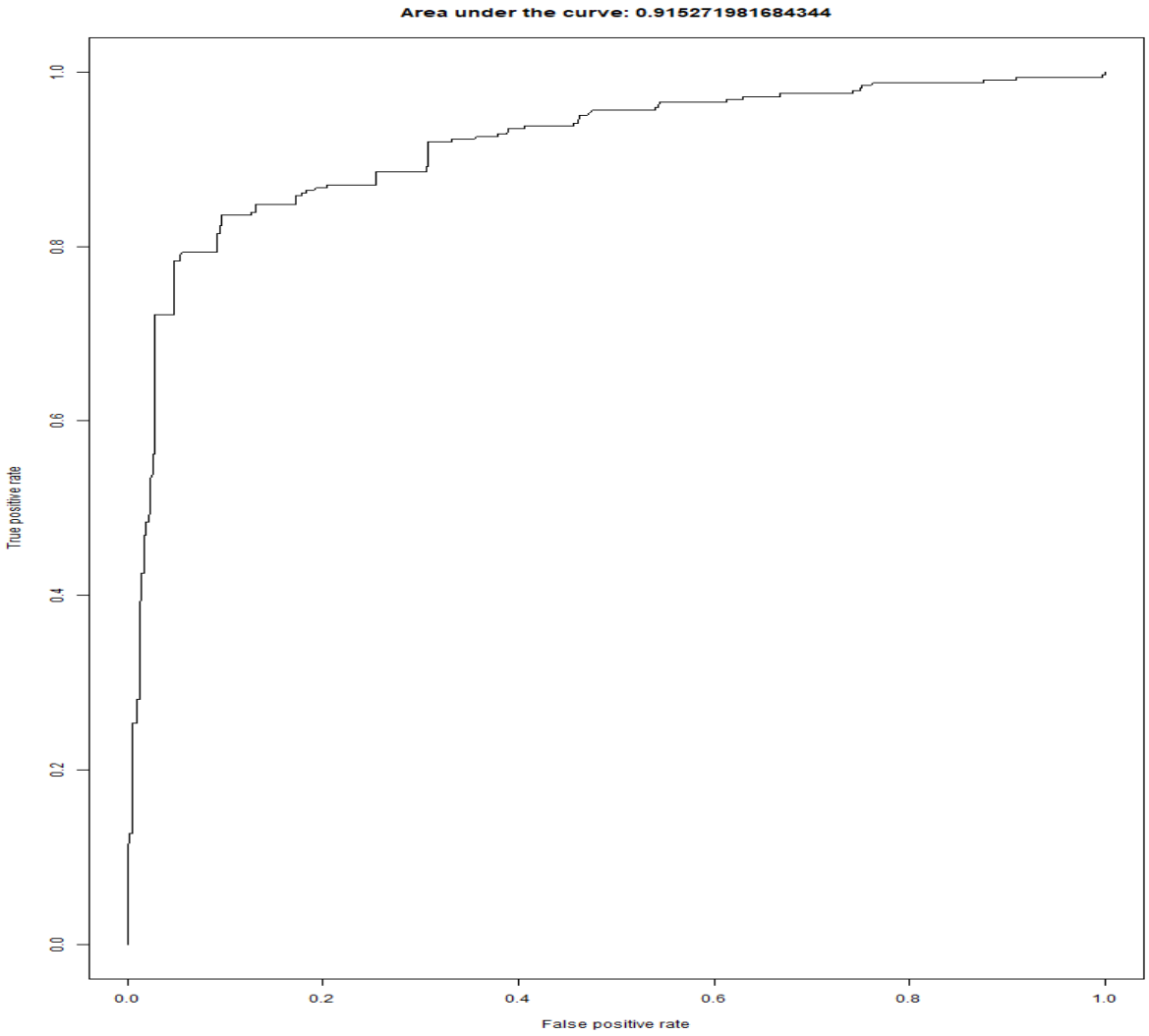
In our case there is 3 categories 10,20,30 the 10 is the reference variable and there is two Boolean variables (price_20, price_30) so if the price value is 10 (price_20 =0 , price_30= 0)

Question5

1. Write the value of AUC.
   0.915272

2. What is the maximum value of AUC (ideal case)?

   The maximum value of AUC is 1, which represents a perfect classifier



Question6

What does each point in the ROC graph represent?
In other words, what is the value that changes and drives TPR and FPR to change too from one point to another in the graph?

Each point on the ROC curve represents a different classification performance at a specific threshold.
To control the TPR and the FPR use the threshold as Changing the threshold changes both:

- True Positive Rate (TPR)
- False Positive Rate (FPR)

Question7

How is the predicted probability affected by changing only Price holding all other variables constant?

As the price increases while the other features are constant the purchase probability decreases.

	Income	Age	Price	PurchaseP
1	42.492	35.976	10	0.6707408
2	42.492	35.976	20	0.4918407
3	42.492	35.976	30	0.1826131

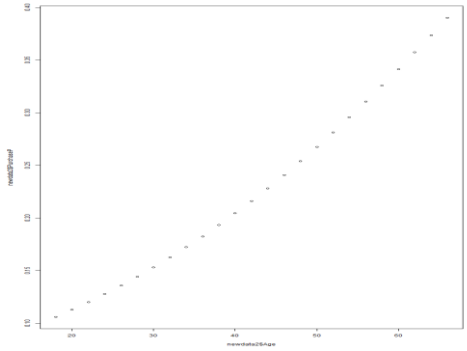
Question8

How is the predicted probability affected by changing only Age holding all other variables constant?

As the Age Increases while other features are constant

the purchase probability increases this means

that there is a direct proportion between them.



Question 9

How is the predicted probability affected by changing only Income holding all other variables constant?

As the Income Increases while other features are constant the purchase probability increases this means that there is a direct proportion between them.

